

Biomechanical Analysis of Ki-Gong Meditation Effects on Center of Mass Acceleration Jerk Variability During Tang Soo Do Training

James Sypherd

Introduction:

While Motion Capture technology is the backbone of video game development, the value of Motion Capture goes beyond the development of animated fantastical creatures. Motion Capture is a tool for biomechanical analysis of complex activities for years utilizing inverse kinematics. The data acquired in the analysis of complex activities has the potential to benefit athletes in their understanding of how to best maximize their Center of Mass Acceleration and improve their performance and to understand what might hinder or lead to adverse effects when the Center of Mass Acceleration is not properly taken into consideration.

The concept of Motion Capture is simple; roughly 20-30 infrared (IR) cameras are placed around a room, objects or people move in front of them while wearing markers made from IR reflecting material. The IR light reflected by these markers is then picked up by the cameras and recreated in a digital 3D space. Here the positional and movement data of these markers can be assigned joints, object names, and attributes thus allowing for replicas to be constructed. See Figure 1:

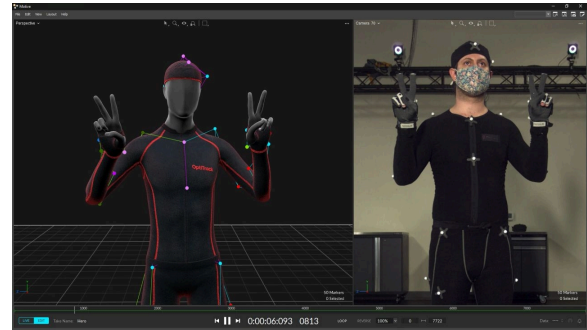


Figure 1: Optitrack Motion Capture in Use [1]

The markers can be placed at specific joints, and from those placements; distance, acceleration, and velocity of the body can be calculated. These values can then be used in inverse kinematics, to calculate mass acceleration, work done, and momentum.

Motion Capture as a tool has been used in academia previously to quantify many different aspects of martial arts. For example, Szczesna et al. used Motion Capture Technology in multiple Kyokushin karate athletes performing the same 4 techniques. [2]

For this project, a similar dataset was collected from ASU students enrolled in AME 294: Martial Arts and Motion Capture. These students were taught four weeks of Tang Soo Do with half the group learning Ki-Gong as an intervention. The goal of this study was to take these new individuals, undeveloped in the Korean martial art of Tang Soo Do, and track their Center of Mass

Acceleration Variability using Motion Capture technology. This was done to four individuals over the span of a session B semester, March to April.

Center of Mass Acceleration Jerk Variability (COMA-JV) is the measurement of the change in tangential acceleration, which happens during voluntary movement. However, the lower this value is, the smoother the movement. In this study, an experimental group of two people had Ki-Gong practices as a part of their regular coursework, which was then contrasted with a control group consisting of two participants who did not.

Their proficiency was charted over 3 weeks, using the first form from the World Tang Soo Do association: “Sae Kye Hyung Il Bu” [3] with and without a ten minute Ki-Gong meditation session. Sae Kye Hyung Il Bu (SKH for short) covers the most basic of concepts a martial artist should be expected to know. These include basic hand and foot techniques, as well as two out of the three core stances.

In this study, movements by Master Ian Larson were considered as the gold-standard, as he has over 25 years of martial arts experience and possesses a 4th Degree Master Rank in Tang Soo Do.

Using the data collected, 95% confidence intervals were calculated to see if Ki-Gong meditation lowered COMA-JV through determining statistical significance. Additionally gold-standard values were compared to each group’s 95% CI to determine if either approached it in three weeks.

Materials and Methods:

Materials

All participants completed informed consent prior to study enrollment. Motion Capture suits and markers were sourced through the ASU Arts and Media Engineering Department (AME). The suits and markers are OptiTrack branded and have been accumulated over the 15 years that the motion capture studio has been operational. Forty-nine markers were used following the setup provided by Optitrack for collection of baseline and passive fingers. [4] MOCAP data was procured using an OptiTrack system with 16 cameras that are a mix of 11 Prime13 and 5 PrimeX13 cameras running at 120 frames per second. Motive version 3.1.1 was the software used to collect data. Data processing was performed using Matlab suite R2024a after collected data sets were exported as CSV files.

Martial Arts Instruction

Participants were instructed in the basics of Tang Soo Do and Ki-Gong depending on assigned group participation during class time in the aforementioned AME 294 class.

The critical components of this instruction were the first Ki-Gong and Tang Soo Do forms, known as Choon Bee Woon Dong (CBW) [5] and Sae Kye Hyung Il Bu (SKH) respectively. The former is a preset list of “soft style” meditative moves and breathing exercises that takes about ten minutes to complete. This form emphasizes relaxed positioning and bodily awareness

which are believed to have lasting effects that decrease COMA-JV.

Sae Kye Hyung Il Bu is a similar set of movements, however it should only take 30 seconds to complete. This preset form utilizes “hard style” martial arts which consists of closed hand strikes, forceful kicks, and strong foot positioning.

Forms like these are designed for the practice of effective transitions between techniques. This allows for better continuity and flow during active combat.

Data Collection

Data collection for the experimental groups was as follows:

- 10 minutes Ki-Gong meditation of CBW
- 10 seconds of standing postural sway with MOCAP tracking
- 45 seconds of Sae Kye Hyung Il Bu (1st martial arts form) with MOCAP
- 10 seconds of standing postural sway with MOCAP tracking
- 10 minutes Ki-Gong meditation

The Ki-Gong meditation was administered through the usage of a prerecorded video that participants followed along with. The purpose of doing this is to hopefully induce a meditative state that will result in less COMA-JV than the control group. Ten seconds of standing postural sway was then collected out of necessity as a baseline for kinematic analysis. SKH was then performed by the participants for a total of 45 seconds to ensure uniform data acquisition. The session was then ended with ten more seconds of standing postural sway and 10 minutes of CBW for insurance.

Data collection for the control group was as follows:

- 10 minutes of no ki-gong - watching The Tonight Show with Jimmy Fallon
- 10 seconds of standing postural sway with MOCAP tracking
- 45 seconds of Sae Kye Hyung Il Bu (1st martial arts form) with MOCAP
- 10 seconds of standing postural sway MOCAP tracking
- 10 minutes of no ki-gong - watching Jimmy Fallon, etc.

Data collection followed the same form as with the experimental group. The only change was that instead of ten minutes of meditation, participants were asked to watch 10 minutes of the Tonight Show. This was done to ensure that participants in this group were not in a meditative state.

Data from both groups was then exported as a CSV and stored within a secure Google Drive. Personal characteristics necessary for biomechanical calculations were recorded and stored in the same Drive. These included height, weight, age, and sex.

Data Processing

CSV's of patient data were imported into the Matlab Workspace wherein the location of the hip markers was extracted, and the centroid was calculated using the following equations:

$$WaistBack_C = \frac{WaistLBack + WaistRBack}{2}$$

$$WaistFront_C = \frac{WaistLFront + WaistRFront}{2}$$

$$WaistBack_C_r = \sqrt{\sum (WaistBack_C^2)}$$

Where WaistBack_C is the midpoint of the rear waist markers, WaistFront_C is the midpoint front waist markers, and WaisteBack_C_r is the magnitude of the back waist midpoint vector.

From this centroid, acceleration was calculated by taking the double derivative of the positional data at all times. Jerk was then calculated by taking the derivative of acceleration with respect to time:

$$Jerk = \frac{acceleration}{dt}$$

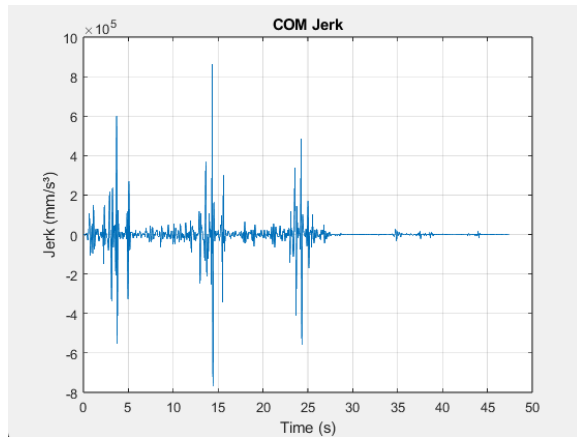


Figure 2: COMA Jerk

The absolute value of which was then taken in order to create a comparable scale value for all data sets to be marked on:

$$total_jerk_magnitude = |Jerk|$$

(see Figure 3)

From here, the Area Under the Jerk Curve was found using a prebuilt MATLAB function that does integration through Cumulative Trapezoidal approximation.

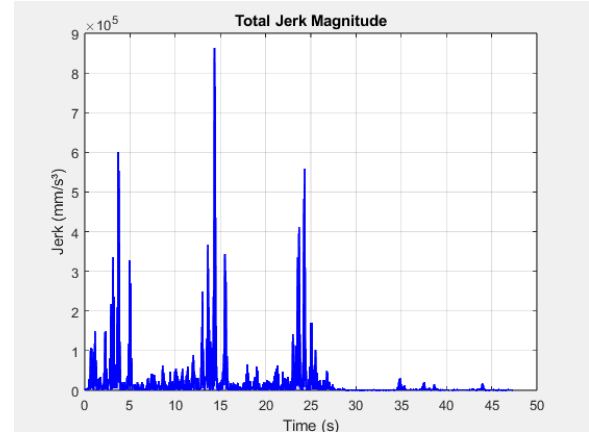


Figure 3: Total Jerk Magnitude

In case this value was not robust enough due to study limitations, the average of the three highest peaks from the Total Jerk Magnitude Chart was found using the FindPeaks Function in MATLAB:

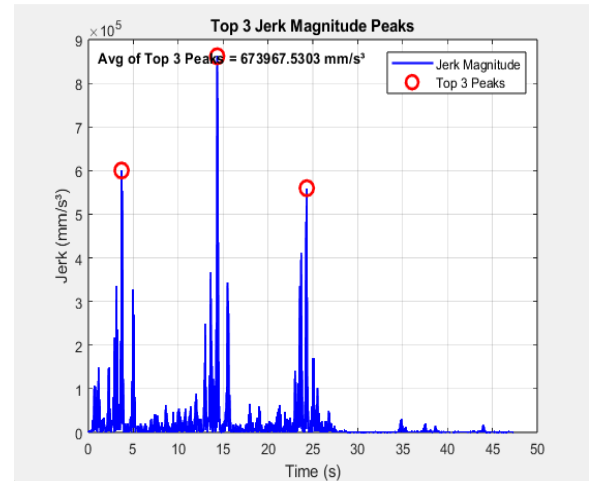


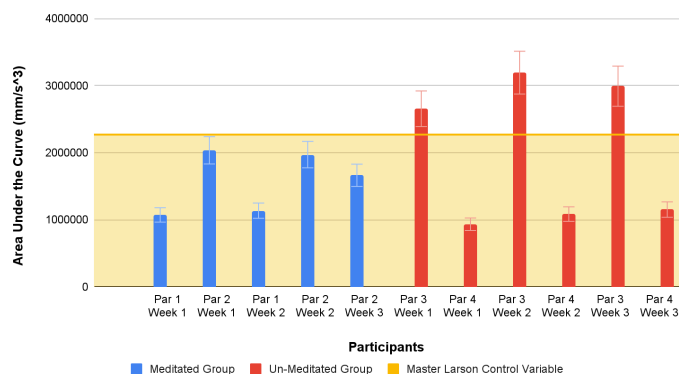
Figure 4: 3 Highest Peaks in Total Jerk Magnitude

Lastly, confidence intervals for each group were calculated in MATLAB. This was done by calculating participant means, standard error, and by using a t-distribution with $n=2$. Margin of error was derived from the t-critical values being multiplied by the standard error of the mean. For the

participant who withdrew in the Meditated Group, the average of their two scores was used. Master Larson's Control Variables were then compared against the confidence intervals to assess efficacy.

Results:

Area Under the Curve



Figures 5: Area Under the Curve Results for All Groups

Data collection across the three weeks went smoothly with only one student withdrawing after 2 weeks due to scheduling conflicts.

For the gold-standard Area Under the Magnitude Curve was recorded at 2,269,610 mm/s³ while his Three Peak Average was recorded to be 811,320 mm/s³.

Participants in the Meditated Group (Group 1) for the first week had Area Under the Magnitude Curve values recorded at 1,072,280 mm/s³ and 2,035,426 mm/s³. While their Three Peak Average was recorded to be 437,936 mm/s³ and 1,031,149 mm/s³. For week two these values shifted to 1,135,355 mm/s³, 1,971,647 mm/s³, 694,208 mm/s³ and 1,206,519 mm/s³ for each of the two previously mentioned categories. Lastly, for week three after participant 1 withdrew,

the values recorded for participant 2 were 1,662,609 mm/s³ and 934,157 mm/s³

For week one, the Un-Meditated Group's (Group 2) values clocked in at 2,654,801 mm/s³ and 933,200 mm/s³ for Area Under the Curve and 1,227,069 mm/s³

Three Peak Average

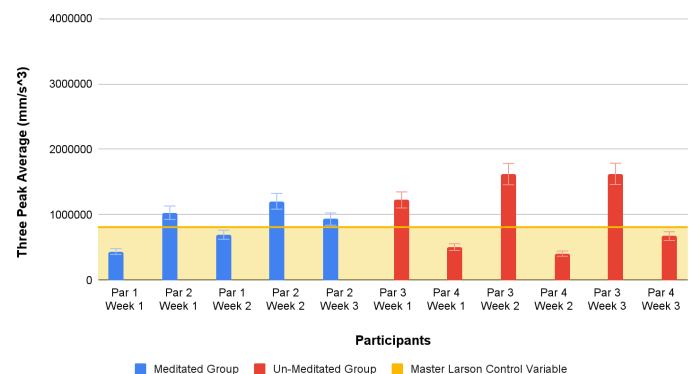


Figure 6: Three Peak Average Results for All Groups

and 505,394 mm/s³ for the Three Peak Average. For week two these values shifted to 3,194,220 mm/s³ and 1,084,284 mm/s³ for Area Under the Curve. Three Peak Average values for week two were 1,621,792 mm/s³ and 405,486 mm/s³. Finally, for week three values for Area Under the Curve were 2,991,934 mm/s³ and 1,152,030 mm/s³ while Three Peak Average values were 1,625,202 mm/s³ and 673,968 mm/s³.

95% Confidence intervals (CI) for Area Under the curve are as follows: Meditated CI = [-3,497,168.48, 6,490,880.19], Un-Meditated CI = [-1,289,103.09, 5,292,593.24]. 95% Confidence intervals for Three Peak Average are: Meditated CI = [-2,308,988.44, 3,932,335.97], Un-Meditated CI = [-5108674.54, 7128311.75].

Discussion:

To reiterate, the main hypothesis of this study was: “Ki-Gong training in conjunction with Tang Soo Do training results in lower COMA-JV” Since all of the calculated confidence intervals cross zero and include the gold-standard, data showed no statistical significance between the three groups.

So why is this? As seen in the graphs above, Master Larson’s Area Under the Curve is consistently higher than the participants from both groups, excluding participant three (this will be discussed later). His Three Peak Average on the other hand tends to be in between the recorded values for each participant. This implies that an experienced Tang Soo Do practitioner does not have bursts of COMA-JV but rather maintains a fairly constant COMA-JV for each movement. The basis of this statement is that for a low average value, but a high total value to be possible, the participants rate has to be kept fairly constant. This contradicts what we went into the study expecting, i.e. that a better martial artist has a lower COMA-JV value and instead offers the idea that the goal is to be somewhere in the middle.

Contrasting the two groups shows that the Meditated Group had, on average, lower values than the Un-Meditated Group with the outlier being participant three. Unlike the other participants who had minimal martial arts experience, participant three is heavily experienced in XMA and American Kenpo fighting styles. Even going so far as to compete in North American

Sports Karate Association events (NASKA). This participant's training emphasized small jumps when transitioning stances that would result in a higher COMA-JV. That being said this participant had no experience with Tang Soo Do and learned at the same rate as the rest of the class. They simply had some habits that affected their data.

If this study were to be redone, more complex biomechanics would be used as this study relied on fairly simple centroid calculations. Additionally more participants with less martial arts experience would be enrolled to limit statistical variability and improve result clarity. More researchers should be added to simplify data collection and a different motion capture system with increased power would be used to improve result specificity. Ideally, the duration of the study would be increased to 6-12 months to try and see truly marketed changes. Lastly, the hypothesis would change from minimizing COMA-JV to instead reaching a middle value possibly based on the average values of multiple masters of Tang Soo Do.

Conclusion:

Biomechanical quantification of improvements over the course of martial arts training is extremely doable using motion capture systems. However in this study no statistical significance was found. For Tang Soo Do practitioners, a low COMA-JV does not seem to be ideal. Rather a constant COMA-JV with low extrema appears to be the goal based on data collected from a 25-year experienced martial artist.

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Works Cited:

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[3] *Sae Kye Hyung Il Bu*. YouTube. (n.d.-b).
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